

# Giovanni Fereoli, M.S.

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## EDUCATION

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- **Ph.D. Aerospace Engineering, University of Colorado Boulder** January 2024 – Present  
*Track: Astrodynamics and Satellite Navigation Systems* *GPA: 4.00/4.00*
  - **Research:** Development of advanced navigation and orbit determination methods for interplanetary and small-body missions, combining interior gravity modeling, Bayesian Markov Chain Monte Carlo inference, map-free Terrain-Relative Navigation, shape reconstruction, small-forces estimation, and machine learning techniques.
  - **Advisor:** Dr. Jay W. McMahon, ORCCA Laboratory.
- **M.S. Space Engineering, Politecnico di Milano** September 2021 – December 2023  
*Focus: Spacecraft GNC, Astrodynamics, Estimation Theory, Numerical Methods* *Grade: 110/110 cum Laude*
  - **Thesis:** Meta-Reinforcement Learning for Spacecraft Proximity Operations Guidance and Control in Cislunar Space. Published findings in the *Journal of Spacecraft and Rockets*.
  - **Advisor(s):** Dr. Pierluigi Di Lizia, SPIRE Laboratory. Dr. Hanspeter Schaub, AVS Laboratory.
- **B.S. Aerospace Engineering, Politecnico di Milano** September 2018 – July 2021  
*Focus: Calculus, Control Theory, Aerodynamics, Structural Mechanics* *Grade: 107/110*
- **Liceo Scientifico** 2013 – 2018  
*Focus: Mathematics, Physics, and Natural Sciences* *Grade: 98/100*

## EXPERIENCE

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- **Visiting Researcher, RSPE Laboratory** University of Bologna  
*Spacecraft Navigation Engineer* *Forthcoming: February 2027 – June 2027*
  - **Research Contribution:** Planned research collaboration on the Radio Science Experiment (RSE) for the Hera mission at Didymos and Dimorphos, with focus on nonlinear estimation methods, near-surface gravity modeling, particle tracking, and optical navigation.
- **Flight Dynamics Affiliate, LASP** University of Colorado Boulder  
*Spacecraft Navigation Engineer* *January 2024 – Present*
  - **Mission Navigation Operations:** Supported the navigation system for the Emirates Mission to the Asteroid Belt, contributing to Mission CDR and GSCDR reviews and developing functional and training exercises for the UAE national mission team.
  - **Scarabaeus Orbit Determination Software Development:** Developer of *Scarabaeus*, an interplanetary orbit determination and navigation software framework. Implemented and maintained high-fidelity dynamics and variational equations, radiometric and optical measurement models with analytical sensitivities, and estimation pipelines for spacecraft states and model parameters. Contributed to supporting infrastructure in Python and Rust, including verification & validation (V&V), unit testing, and development of a MongoDB-based mission database, with 800+ commits across the codebase.
- **Visiting Researcher, AVS Laboratory** University of Colorado Boulder  
*Machine Learning Engineer* *March 2023 – December 2023*
  - **Research Contribution:** Conducted research on the application of Reinforcement Learning for spacecraft guidance and control, focusing on proximity operations between spacecraft.

## SKILLS

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**Languages:** Python (NumPy, SciPy, TensorFlow, PyTest, Matplotlib), C++ (Eigen), Rust, MATLAB, Julia, LaTeX, AMPL

**Astrodynamics Software:** SPICE Toolkit, GMAT, MONTE, SBMT

**Tools & Platforms:** Git, Linux/Unix, Bash, AWS EC2, MongoDB, Atlassian Suite (Confluence, Jira, Bitbucket)

**Engineering Software:** SolidWorks, Solid Edge, Autodesk Inventor, Blender

## HONORS & AWARDS

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- **Excellent Reviewer Recognition – Journal of Guidance, Control, and Dynamics (JGCD):** Selected as an Excellent Reviewer for the period 10/1/2024–9/30/2025 in recognition of outstanding contributions to the peer-review process (January 2026).
- **John V. Breakwell Student Award:** Awarded at the 2025 AAS/AIAA Astrodynamics Specialist Conference in Boston, MA, for outstanding student research in astrodynamics (August 2025).
- **Europe-Colorado Program Scholarship:** Awarded by the University of Colorado Boulder College of Engineering & Applied Science (April 2022).

## LANGUAGES SPOKEN

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- **English:** Full Professional Proficiency (Speaking, Reading, Writing)
- **French:** Memorized Proficiency (Speaking, Reading, Writing)
- **Italian:** Native (Speaking, Reading, Writing)

## LICENSES & CERTIFICATIONS

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- **C++ for Programmers Course:** Codecademy, Issued Jan 2024 (Credential ID: 659C8613AB)
- **Dynamics and Real Time Simulation (DARTS) Lab Course:** NASA Jet Propulsion Laboratory, Issued Aug 2023
- **Learn Python 3 Course:** Codecademy, Issued Sep 2022 (Credential ID: 631F07BB45)
- **TOEIC (985/990):** ETS, Issued Aug 2021
- **Space Exploration: From Past to Future:** Politecnico di Milano, Issued Dec 2019
- **BLS-D (Basic Life Support and Defibrillation):** Croce Rossa Italiana, Issued Mar 2017

## JOURNAL PAPERS

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- **Meta-Reinforcement Learning for Spacecraft Proximity Operations Guidance and Control in Cislunar Space:** Giovanni Fereoli, Hanspeter Schaub, Pierluigi Di Lizia, *Journal of Spacecraft and Rockets*, October 2024, DOI: 10.2514/1.A36100
- **Interior Gravity Characterization of Small Celestial Bodies Using Cylindrical Harmonics:** Giovanni Fereoli, Jay McMahon, *Celestial Mechanics and Dynamical Astronomy*, March 2026, DOI: 10.1007/s10569-026-10281-7
- **Bayesian Orbit Determination via Markov Chain Monte Carlo with High-Order Flow Expansions:** Giovanni Fereoli, Jay McMahon, *Acta Astronautica*, in preparation
- **REVO: Rotation Estimation and Visual Odometry for Surface Relative Navigation:** J. Villa, Giovanni Fereoli, Jay McMahon, *Journal of Guidance, Control, and Dynamics*, in preparation
- **Small-Body Interior Density Inference and Localized Mass Variation Reconstruction via Cylindrical Harmonic Gravity Modeling:** Giovanni Fereoli, Jay McMahon, *Icarus*, in preparation

## CONFERENCE PAPERS

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- **Meta-Reinforcement Learning for Spacecraft Proximity Operations Guidance and Control in Cislunar Space:** Giovanni Fereoli, Hanspeter Schaub, Pierluigi Di Lizia, presented at the *46th AAS Guidance, Navigation, and Control Conference*, Breckenridge, Colorado, February 2024.
- **Design and Preliminary Results of Scarabaeus: A New Open-Source Navigation Tool for Interplanetary Spacecraft Navigation:** Jay McMahon, Mattia Pugliatti, Dahlia Baker, Anivid Pedros-Faura, Giovanni Fereoli, Kian Shakerin, Santhosh Pattamudu-Manoharan, Zachary Ellis, Mohamed Almashjari, Mohamed Kuleib, Wendy Frank, Jacopo Villa, Jeremy Knittel, presented at the *47th Rocky Mountain AAS GN&C Conference*, Breckenridge, Colorado, February 2025.
- **On Cylindrical Harmonics for Local Gravity Field Modeling:** Giovanni Fereoli, Jay McMahon, presented at the *2025 AAS/AIAA Astrodynamics Specialist Conference*, Boston, Massachusetts, August 2025.
- **Operational Maneuver Targeting Framework for the Emirates Mission to the Asteroid Belt:** Mohamed Kuleib, Giovanni Fereoli, Jay McMahon, Jeremy Knittel, *30th International Symposium on Space Flight Dynamics (ISSFD)*, Toulouse, France, June 2026

- **The Scarabaeus Open-Source Navigation Tool: Preliminary Results and Real Measurements:** Jay McMahon, Mattia Pugliatti, Giovanni Fereoli, Santhosh Pattamudu-Manoharan, Zachary Ellis, Annalise Cabra, Mohamed Almashjari, Mohamed Kuleib, Wendy Frank, Jeremy Knittel, presented at the *2025 AAS/AIAA Astrodynamics Specialist Conference*, Boston, Massachusetts, August 2025.
- **Efficient Posterior Sampling for Small-Body Orbit Determination:** Giovanni Fereoli, Jay McMahon, manuscript in preparation for the *2026 AAS/AIAA Astrodynamics Specialist Conference*, Whistler, British Columbia, Canada, August 2026.

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## WORKSHOPS

- **Reinforcement Learning for Proximity Operations Applications:** Giovanni Fereoli presented at the *8th AIAA Intelligent Systems Workshop*, Boulder, Colorado, July, 2023

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## PROFESSIONAL SERVICE

- **Peer Reviewer:** Served as a peer reviewer for the following journals:
  - *AIAA Journal of Guidance, Control, and Dynamics* — 1 manuscript reviewed
  - *Acta Astronautica* — 6 manuscript reviewed
  - *Journal of Open Source Software (JOSS)* — 1 software package reviewed: `JS0Solvers.jl`

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## SUPERVISION

- **Claudia Muñoz (M.S. Student, Technische Universität München):** Visiting student at University of Colorado Boulder. Supervised her project *“Deep Learning-Based Online Density Adaptation for Aerocapture Guidance”* (2025).

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## MEDIA MENTIONS

- **Radio Duchessa Interview:** Interviewed about space engineering and research journey on Radio Duchessa (2025).